DO YOU KNOW?

Relative Molecular Mass is only a ratio. So, it has no unit. If the molecular mass of a compound is expressed in grams, it is called ***Gram Molecular Mass.***

Gram Molecular Mass of water	=	18 g
Gram Molecular Mass of carbon dioxide	=	44 g
Gram Molecular Mass of ammonia	=	17 g
Gram Molecular Mass of HCl	=	36.5 g

The relative molecular mass is obtained by adding together the relative atomic masses of all the atoms present in a molecule.

Calculation of relative molecular mass – Solved examples:

Example 1: Relative molecular mass of sulphuric acid (H_2SO_4) is calculated as follows: Sulphuric acid conatins 2 atoms of hydrogen, 1 atom of sulphur and 4 atoms of oxygen.

Therefore, Relative molecular mass of sulphuric acid = (2 × mass of hydrogen) + (1 × mass of sulphur) + (4 × mass of oxygen)

$= (2 \times 1) + (1 \times 32) + (4 \times 16)$

$= 98$

i.e., one molecule of H_2SO_4 is 98 times as heavy as $\frac{1}{12^{th}}$ of the mass of a carbon –12.

Example 2: Relative molecular mass of water (H_2O) is calculated as follows: A water molecule is made of 2 atoms of hydrogen and one atom of oxygen.

So, the relative molecular mass of water

= (2 × mass of hydrogen) + (1 × mass of oxygen)

$= (2 \times 1) + (1 \times 16)$

$= 18$

i.e., one molecule of H_2O is 18 times as heavy as $\frac{1}{12^{th}}$ of the mass of a carbon –12.

7.3 DIFFERENCE BETWEEN ATOMS AND MOLECULES

Even though atoms are the basic components of molecules, they differ in many aspects when compared to the molecules. Table 7.5 consolidates the major difference between atoms and molecules.

Table 7.5 Difference between atoms and molecules

Atom	Molecule
An atom is the smallest particle of an element	**A molecule is the smallest particle of an element or compound.**
Atom does not exist in free state except in noble gas	**Molecule exists in a free a state**
Except some of noble gas, other atoms are highly reactive	**Molecules are less reactive**
Atom does not have a chemical bond	**Atoms in a molecule are held by chemical bonds**

7.4 MOLE CONCEPT

So far we discussed about matters in terms of individual atoms and molecules. Atomic mass units provide a relative scale for the masses of the elements. Since the atoms have such small masses, no usable scale can be devised to weigh them in the calibrated units of atomic mass units. In any real situation, we deal with macroscopic samples containing enormous number of atoms. Therefore, it is convenient to have a special unit to describe a very large number of atoms. The idea of a 'unit' to denote a particular number of objects is not new. For example, **the pair (2 items) and the dozen (12 items)**, are all familiar units. Chemists measure atoms and molecules in 'moles'. So, you can now understand that 'mole' denotes a number of particles.

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